

# An Improved Differential Evolution-Based Hand-Eye Calibration Optimization Algorithm with Elastic Net Regularization: Supplementary File

## I. ADDITIONAL TABLES

TABLE S.I  
JOINT ANGLES OF THE INDUSTRIAL ROBOT MOVEMENT

| No. | $q_1$  | $q_2$  | $q_3$ | $q_4$ | $q_5$ | $q_6$ | No. | $q_1$  | $q_2$  | $q_3$ | $q_4$ | $q_5$ | $q_6$ |
|-----|--------|--------|-------|-------|-------|-------|-----|--------|--------|-------|-------|-------|-------|
| 1   | -10.52 | -8.05  | -6.52 | -1.60 | 89.55 | -4.10 | 21  | -11.40 | -11.88 | -6.52 | -5.72 | 92.79 | -4.10 |
| 2   | -16.48 | -8.05  | -6.52 | -1.60 | 89.55 | -4.10 | 22  | -11.40 | -11.88 | -6.52 | -5.72 | 88.61 | -4.10 |
| 3   | -13.21 | -7.52  | -6.52 | -1.60 | 89.55 | -4.10 | 23  | -11.40 | -11.88 | -6.52 | -7.77 | 88.61 | -4.10 |
| 4   | -8.24  | -7.52  | -6.52 | -1.60 | 89.55 | -4.10 | 24  | -11.40 | -11.88 | -6.52 | 2.35  | 88.61 | -4.10 |
| 5   | -5.32  | -7.52  | -6.52 | -1.60 | 89.55 | -4.10 | 25  | -11.40 | -11.88 | -6.52 | 2.35  | 87.42 | -4.10 |
| 6   | -4.20  | -7.62  | -6.52 | -1.60 | 89.55 | -4.10 | 26  | -11.40 | -11.88 | -6.52 | 2.35  | 92.60 | -4.10 |
| 7   | -3.88  | -9.90  | -6.52 | -1.60 | 89.55 | -4.10 | 27  | -11.40 | -11.88 | -6.52 | 4.17  | 92.60 | -4.10 |
| 8   | -10.18 | -9.90  | -6.52 | -1.60 | 89.55 | -4.10 | 28  | -11.40 | -11.88 | -6.52 | 4.17  | 93.63 | -4.10 |
| 9   | -13.68 | -9.90  | -6.52 | -1.60 | 89.55 | -4.10 | 29  | -11.40 | -11.88 | -6.52 | 4.17  | 89.49 | -4.10 |
| 10  | -16.51 | -11.88 | -6.52 | -1.60 | 89.55 | -4.10 | 30  | -11.40 | -11.88 | -6.52 | -1.48 | 89.49 | -4.10 |
| 11  | -11.40 | -11.88 | -6.52 | -1.60 | 89.55 | -4.10 | 31  | -11.40 | -11.88 | -6.52 | -1.48 | 93.91 | -4.10 |
| 12  | -11.40 | -11.88 | -6.52 | -1.60 | 93.12 | -4.10 | 32  | -11.40 | -11.88 | -6.52 | -6.45 | 93.79 | -4.10 |
| 13  | -11.40 | -11.88 | -6.52 | 2.79  | 93.12 | -4.10 | 33  | -11.40 | -11.88 | -6.52 | -6.45 | 88.66 | -4.10 |
| 14  | -11.40 | -11.88 | -6.52 | -4.13 | 93.12 | -4.10 | 34  | -11.40 | -11.88 | -6.52 | 2.34  | 88.66 | -4.10 |
| 15  | -11.40 | -11.88 | -6.52 | -4.13 | 88.56 | -4.10 | 35  | -11.40 | -11.88 | -6.52 | 2.34  | 94.15 | -4.10 |
| 16  | -11.40 | -11.88 | -6.52 | -6.95 | 88.56 | -4.10 | 36  | -11.40 | -11.88 | -6.52 | 4.69  | 93.65 | -4.10 |
| 17  | -11.40 | -11.88 | -6.52 | 1.67  | 88.56 | -4.10 | 37  | -16.25 | -11.43 | -6.52 | 4.69  | 93.65 | -4.10 |
| 18  | -11.40 | -11.88 | -6.52 | 1.67  | 92.98 | -4.10 | 38  | -18.82 | -11.43 | -6.52 | 4.69  | 93.65 | -4.10 |
| 19  | -11.40 | -11.88 | -6.52 | 4.23  | 92.98 | -4.10 | 39  | -22.21 | -11.45 | -6.52 | 4.69  | 93.65 | -4.10 |
| 20  | -11.40 | -11.88 | -6.52 | -0.19 | 92.98 | -4.10 | 40  | -24.75 | -11.58 | -6.52 | 4.69  | 93.65 | -4.10 |

TABLE S.II  
ALGORITHM PARAMETER SETTINGS

| Algorithm | Parameter Settings                                                                                                                     |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------|
| IDER      | $N_P = 100$ ; $T = 120$ ; Scaling factor $F = 0.6$ ; Crossover rate $CR = 0.9$ .                                                       |
| PSO       | $N_P = 100$ ; $T = 120$ ; Inertia weight $\omega = 0.6$ ; Learning factors $c_1 = 1.2$ , $c_2 = 1.2$ .                                 |
| DE        | $N_P = 100$ ; $T = 120$ ; Scaling factor $F = 0.6$ ; Crossover rate $CR = 0.9$ .                                                       |
| GA        | $N_P = 100$ ; $T = 120$ ; Crossover rate: 0.9; Mutation rate: 0.08; Tournament selection size: 3.                                      |
| QPSO      | $N_P = 100$ ; $T = 120$ ; Contraction-expansion coefficient: 1; Mean optimal coefficient: 0.06.                                        |
| CS        | $N_P = 100$ ; $T = 120$ ; Abandonment probability $pa = 0.15$ ; Step size scaling factor: 0.6; Characteristic exponent $\beta = 1.8$ . |
| GWO       | $N_P = 100$ ; $T = 120$ ; Initial convergence factor: 1.8; Final convergence factor: 0.                                                |
| SCA       | $N_P = 100$ ; $T = 120$ ; Initial attenuation factor: 5; Final attenuation factor: 0.                                                  |
| ABC       | $N_P = 100$ ; $T = 120$ ; Limit: 15; Scout bee trigger frequency: 12.                                                                  |
| FA        | $N_P = 100$ ; $T = 120$ ; Maximum attractiveness: 1; Light absorption coefficient: 0.8; Random step factor: 0.7.                       |

This is the supplementary material for the paper. It contains extra tables and figures concerning the model structure and experimental results.

**Algorithm 1** IDER Algorithm.**Input:**  $x_0, A, B$ 

INITIALIZATION:

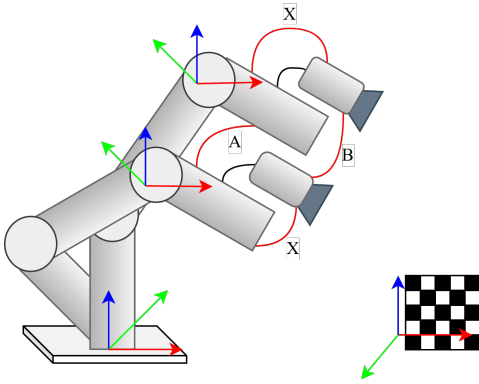
1. **Initialize:**  $F, CR, T, N_P$
2. **Initialize:**  $x_{de} = x_0, f(X_{de}) = f(X_0)$

OPTIMIZATION STEP:

3. **for**  $t = 1$  **to**  $T$
4.     Update  $x_i$  with Eq. (23)
5.     Calculate  $v_i$  with Eq. (26)
6.     Update  $u_i$  with Eq. (28)
7.     Update  $u_i$  with Eq. (30)
8.     Calculate  $f_{ui}$  with Eq. (25)
9.     Update  $x_{(t+1)}$  with Eq. (31)
10. **end for**
11.      $x_{de} = x_{(t+1)}$
- REGULARIZATION STEP:
12.     **for**  $t = 1$  **to**  $T$
13.         **if**  $|E_1| \geq 1$
14.             Update  $P_i^{(t+1)}$  with Eq. (38)
15.         **else**
16.             Update  $P_i^{(t+1)}$  with Eq. (39)
17.         **end if**
18.         Calculate  $F(\alpha_{\text{opt}}, \lambda_{\text{opt}})$  with Eq. (34)
19.         Update  $(\alpha^*, \lambda^*)$  with Eq. (42)
20.         Calculate  $x_{en}$  with Eq. (33)
21.         Update  $X_{en}$
22.         Calculate RMSE with Eq. (43)
23.     **end for**

**Output:**  $X_{en}, \text{RMSE}$ 

## II. ADDITIONAL FIGURES

Fig. S.1. Formulating the  $AX=XB$  problem.